

N-Nitrosamine Evaluation Tools

These modules are designed to evaluate the potential carcinogenic potency of *N*-nitrosamine compounds that have not undergone adequate animal carcinogenicity testing. The primary module for this purpose is the *CPCA Prediction module*, which is designed for assigning *N*-nitrosamine impurities (including NDSRIs) to a predicted carcinogenic potency category and a corresponding acceptable intake (AI) limit. This module also performs sophisticated surrogate (with animal carcinogenicity data) search and finding similar NDSRIs for which regulatory agencies have published acceptable limits (AIs).

The *Surrogate Search* module is the legacy module to search for *N*-Nitrosamine surrogates with available animal carcinogenicity data. This module was introduced few years before the introduction of the CPCA method. It is accompanied by additional modules such as LogP, water solubility, Ames mutagenicity, and the Gold Carcinogenicity Potency Database (CPDB).

Table 1. QSAR Flex models for N-Nitrosamine modules

Model Name	Version	Model Description	Reference Database Size
CPCA Prediction	12.0	Carcinogenic Potency Categorization Approach (CPCA) for <i>N</i> -Nitrosamines	Surrogates = 209 NDSRIs with AI = 577
Surrogate Search	6.0	Analog-based read across using nitrosamine local environment	209

Model Details

CPCA Prediction: This module is based on the Carcinogenic Potency Categorization Approach (CPCA) methodology for *N*-Nitrosamines, which was published by the European Medicines Agency (Q&A) in July 2023 and followed by the US FDA (Guidance) in August 2023. This is now the standard recommended methodology to assign a recommended AI limit to an NDSRI based on the NDSRI's activating and deactivating structural features. This module supports both EMA and US-FDA specifications, automatically counts α -Hydrogens and identifies deactivating and activating features, implements the CPCA flowchart and has sophisticated surrogate and NDSRI search techniques built-in.

Surrogate Search: The carcinogenic potency of nitrosamines varies significantly. Due to the lack of testing data for complex nitrosamines, evaluating their potential impact on health is a challenging task. Computational methods such as surrogate-based SARs have been proposed as alternative approaches. The *Surrogate Search* module performs such a method to search for *N*-Nitrosamine surrogates. By submitting a query nitrosamine, this module retrieves nitrosamine analogs with available carcinogenicity data that share similar structural environment around

the nitrosamine functionality (local similarity) along with other properties such as Ames mutagenicity, water solubility, LogP etc. This information can help experts determine the acceptable intake limit for the query nitrosamine or to uncover important structure-activity relationships.

References

1. Recommended Acceptable Intake Limits for Nitrosamine Drug Substance-Related Impurities (NDSRIs), Guidance for Industry, U.S. Department of Health and Human Services; US FDA; Center for Drug Evaluation and Research (CDER), August 2023.
2. Questions and answers for marketing authorisation holders/applicants on the CHMP Opinion for the Article 5(3) of Regulation (EC) No 726/2004 referral on nitrosamine impurities in human medicinal products, European Medicines Agency, July 2023.
3. Chakravarti, S. K., FlexFilters: A Scalable and Flexible QSAR Platform for Addressing Complex and Diverse Types of in Silico Safety Assessment of Chemicals, Poster, Society of Toxicology Annual Meeting, Anaheim, 2020.
4. Saiakhov, R., Exploring Local Structural Environments of Nitrosamine Analogs to Improve Carcinogenicity Read-Across Assessments, Poster presentation. GTA Meeting May 2023.
5. Chakravarti, Suman. "Computational Prediction of Metabolic α -Carbon Hydroxylation Potential of *N*-Nitrosamines: Overcoming Data Limitations for Carcinogenicity Assessment." *Chemical Research in Toxicology*, 2023, 36, 959–970. <https://doi.org/10.1021/acs.chemrestox.3c00083>.
6. NDSRI AI Limits set by the following agencies:
 - Health Canada - <https://www.canada.ca/en/health-canada/services/drugs-health-products/compliance-enforcement/information-health-product/drugs/nitrosamine-impurities/established-acceptable-intake-limits.html>
 - EMA - <https://www.ema.europa.eu/en/human-regulatory-overview/post-authorisation/pharmacovigilance-post-authorisation/referral-procedures-human-medicines/nitrosamine-impurities#related-documents-19209>
 - FDA - <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cder-nitrosamine-impurity-acceptable-intake-limits>